

Initial Sign Vectors in c_0 Refute a Schur-Generation Question

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Source Question

The source is Ondrej Kurka, *Tsirelson-like spaces and complexity of classes of Banach spaces*, arXiv:1612.09334. Remark 5.9(iii), on page 21, asks:

Does a Banach space X has necessarily the Schur property if it has a subset W such that $\overline{\text{co}}W = B_X$ and every weakly convergent sequence of elements of W is convergent in the norm?

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Remark 5.9. (i) It is possible to use [2, Theorem 3.2] to quantify the Schur property of $E_{\mathcal{T}}^*$ in a bit different direction. If the assumption $\forall \sigma \in 2^{\mathbb{N}} : \mathcal{T}(\sigma) \in \text{IF}$ is met, then $E_{\mathcal{T}}^*$ has the strong Schur property in the sense that every bounded sequence $a_1^*, a_2^*, \dots$ in $E_{\mathcal{T}}^*$ with $\inf\{\|a_n^* - a_m^*\|_{\mathcal{T}} : n \neq m\} > \varepsilon$ has a subsequence $a_{n_j}^*$ such that

$$\left\| \sum_{j=1}^k \lambda_j a_{n_j}^* \right\|_{\mathcal{T}} \geq \frac{\varepsilon}{12} \sum_{j=1}^k |\lambda_j|$$

for all $k \in \mathbb{N}$ and $\lambda_1, \dots, \lambda_k \in \mathbb{R}$.

(ii) If the assumption $\forall \sigma \in 2^{\mathbb{N}} : \mathcal{T}(\sigma) \in \text{IF}$ is met, a quantitative version of the Dunford-Pettis property of $E_{\mathcal{T}}^*$ and of $E_{\mathcal{T}}$ can be obtained as well (see [14, Proposition 6.4 and Theorem 6.5]).

(iii) In [20], a question was considered whether the proposed tree space method can be used for amalgamating of spaces with the Schur property (see [20, Remark 3.7(c)] for the exact formulation). Proposition 5.7(1) shows a concrete example of a family of spaces with the Schur property for which the answer is positive, although not trivial. Let us note that to prove that $E_{\mathcal{T}}^*$ has the Schur property would be much simpler if we had the positive answer to the following question: *Does a Banach space X has necessarily the Schur property if it has a subset W such that $\overline{\text{co}}W = B_X$ and every weakly convergent sequence of elements of W is convergent in the norm?*

Answer

The answer is negative. The counterexample is $X = c_0$, with W the union of all finite initial sign cubes.

Definitions and Normalizations

Let c_0 be the real Banach space of scalar sequences converging to zero, with the supremum norm. The Schur property means that every weakly convergent sequence is norm convergent.

For $n \geq 1$, define

$$W_n = \{(\varepsilon_1, \dots, \varepsilon_n, 0, 0, \dots) : \varepsilon_i \in \{-1, 1\} \text{ for } 1 \leq i \leq n\} \subset c_0$$

and put

$$W = \bigcup_{n=1}^{\infty} W_n.$$

Every element of W belongs to the unit sphere of c_0 .

Proof Intuition

The set W_n is the vertex set of the n -dimensional cube $[-1, 1]^n$, embedded in the first n coordinates of c_0 . Thus the convex hull of W contains every finitely supported point of the unit ball, and these points are dense in B_{c_0} .

The weak-sequential condition holds for a different reason. A weakly convergent sequence in c_0 is coordinatewise convergent. If the sign vectors have unbounded lengths, then after passing to a subsequence every fixed coordinate is eventually a sign, and coordinate convergence forces that coordinate of the weak limit to have modulus one. This would happen for every coordinate, which is impossible for an element of c_0 . So the lengths are bounded, and the sequence lives in a finite set.

Counterexample

Theorem 1. *There is a Banach space X and a subset $W \subset B_X$ such that*

$$\overline{\text{co}}W = B_X,$$

and every weakly convergent sequence of elements of W is norm convergent, but X does not have the Schur property.

Proof. Take $X = c_0$ and $W = \bigcup_{n \geq 1} W_n$ as above.

First we show that $\overline{\text{co}}W = B_{c_0}$. For each fixed n , the convex hull of W_n is exactly the finite cube

$$\{x \in c_0 : \text{supp } x \subseteq \{1, \dots, n\}, |x_i| \leq 1 \text{ for } 1 \leq i \leq n\}.$$

Indeed, this is the usual description of the cube as the convex hull of its 2^n sign vertices. Hence every finitely supported $x \in B_{c_0}$ belongs to $\text{co } W$, after choosing n with $\text{supp } x \subseteq \{1, \dots, n\}$. Since the finitely supported points of B_{c_0} are dense in B_{c_0} by coordinate truncation, $\overline{\text{co}}W = B_{c_0}$.

Now let $(w_j)_j$ be a weakly convergent sequence in W , say

$$w_j \xrightarrow{\text{weak}} x \in c_0.$$

Let n_j be the unique integer such that $w_j \in W_{n_j}$. We claim that the sequence $(n_j)_j$ is bounded. Suppose not. Passing to a subsequence, still denoted $(w_j)_j$, we may assume that $n_j \rightarrow \infty$. For each fixed coordinate k , the coordinate functional $\delta_k \in c_0^* = \ell_1$ gives

$$\delta_k(w_j) \longrightarrow \delta_k(x) = x_k.$$

But for all sufficiently large j , since $k \leq n_j$, we have $\delta_k(w_j) \in \{-1, 1\}$. A convergent scalar sequence whose tail is contained in $\{-1, 1\}$ is eventually constant. Therefore

$$|x_k| = 1$$

for every $k \in \mathbb{N}$. This contradicts $x \in c_0$. Thus the lengths $(n_j)_j$ are bounded.

Choose N such that $n_j \leq N$ for all j . Then

$$\{w_j : j \in \mathbb{N}\} \subseteq \bigcup_{n=1}^N W_n,$$

which is a finite set. Since $(w_j)_j$ is weakly convergent, every subsequence has the same weak limit. A finite-valued convergent sequence in the Hausdorff weak topology must eventually be constant: otherwise two distinct values would occur on infinite subsequences and both would have to equal the weak limit. Hence $(w_j)_j$ is eventually constant, and in particular norm convergent.

Finally, c_0 does not have the Schur property. Let $(e_n)_n$ be the canonical unit vector basis. For every $a = (a_n)_n \in \ell_1 = c_0^*$,

$$a(e_n) = a_n \longrightarrow 0,$$

so $e_n \rightarrow 0$ weakly in c_0 . However, $\|e_n\|_\infty = 1$ for all n , so $(e_n)_n$ is not norm convergent to 0. Therefore c_0 fails the Schur property, while the set W satisfies the two conditions in the source question. $\square$

Verification Notes

The proof is self-contained modulo the standard identification $c_0^* = \ell_1$. The construction also survives harmless variations of the wording in the source question: since $\overline{\text{co}}W = B_X$, any such W must in particular lie inside B_X , and the chosen W already lies in the unit sphere.

The only sequential point is the bounded-length claim. It uses coordinatewise convergence, forced by weak convergence in c_0 , and the fact that a vector whose every coordinate has modulus one cannot belong to c_0 .

Novelty and Duplicate Checks

Before promotion, I searched the run indexes for arXiv 1612.09334, the title terms, **Tsirelson-like spaces**, **Schur property**, **weakly convergent sequence of elements of W** , and related phrases from Remark 5.9(iii). No existing packet for this exact question was found; the nearby hits concerned different Schur, Tsirelson, or Borel complexity questions.

I also ran bounded web searches for exact and near-exact phrases including

- "every weakly convergent sequence of elements of W " "Schur property",
- "Does a Banach space X " "Schur property" "subset W ",
- "Kurka" "subset W " "Schur property",
- " c_0 " "every weakly convergent sequence" "norm convergent" "closed convex hull".

The searches did not surface a prior answer to this embedded question. The packet is therefore marked as a candidate original counterexample pending human review.

References

References

- [1] O. Kurka, *Tsirelson-like spaces and complexity of classes of Banach spaces*, arXiv:1612.09334, 2016.

Human Review Recommendation

This should be reviewed as an elementary counterexample to the displayed question in Remark 5.9(iii). The suggested checks are: verify the transcription of the question, confirm that no additional hidden restriction on W is intended, and inspect the weak-convergence argument for the initial sign-vector set in c_0 . Under the displayed wording, the answer is decisively negative.

Human Review: Initial sign vectors in c_0

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct. It gives a simple counterexample to the question in Remark 5.9(iii) of Kurka's paper, under the statement as printed.

The example is

$$X = c_0,$$

and W is the union of the finite initial sign cubes

$$W_n = \{(\varepsilon_1, \dots, \varepsilon_n, 0, 0, \dots) : \varepsilon_i \in \{-1, 1\}\}.$$

The construction is very natural once the example has been found: the sets W_n are exactly the vertices of the finite cubes in the first n coordinates.

Convex Generation

The verification of

$$\overline{\text{co}}W = B_{c_0}$$

is correct. For each n , the convex hull of W_n is the cube

$$\{x \in c_0 : \text{supp } x \subseteq \{1, \dots, n\}, |x_i| \leq 1\}.$$

Therefore every finitely supported element of B_{c_0} belongs to $\text{co } W$. Since finitely supported elements are dense in B_{c_0} by coordinate truncation, the closed convex hull of W is the whole unit ball.

Weakly Convergent Sequences in W

The weak-sequential part is also correct. Suppose that

$$w_j \in W_{n_j}$$

is weakly convergent in c_0 . Weak convergence in c_0 implies coordinatewise convergence, because the coordinate functionals belong to $c_0^* = \ell_1$.

If the lengths n_j were unbounded, one could pass to a subsequence with $n_j \rightarrow \infty$. Then for every fixed coordinate k , the k -th coordinate of w_j is eventually always a sign, hence lies in $\{-1, 1\}$. Since this coordinate sequence converges, it must eventually be constant, and the weak limit would have

$$|x_k| = 1$$

for every k . This is impossible for $x \in c_0$.

Thus the lengths n_j are bounded. Consequently the sequence (w_j) takes values in a finite set. A finite-valued weakly convergent sequence in a Hausdorff topology is eventually constant, hence norm convergent.

Failure of the Schur Property

Finally, c_0 does not have the Schur property. The unit vector basis (e_n) is weakly null in c_0 , since for every $a = (a_n) \in \ell_1 = c_0^*$,

$$a(e_n) = a_n \rightarrow 0.$$

But $\|e_n\|_\infty = 1$ for all n , so (e_n) is not norm null.

Review Outcome

The proof appears correct. The construction is elementary, and the idea is clear once one notices the right set W : use initial sign vectors to generate the unit ball of c_0 , while forcing weakly convergent sequences in W to have bounded length and hence to be eventually constant.

This should be recorded as an apparently correct counterexample to the question as printed, subject only to the usual semantic check that no extra condition on W was intended.